

CPMS R&D; Program – Automatic Number Plate Recognition (ANPR) Research Initiative

Background

The Car Park Management System (CPMS) platform is continually evolving to improve automation, operational efficiency, security, and customer experience across parking facilities.

One of the strategic enhancements under evaluation as part of the CPMS research and development roadmap is the integration of Automatic Number Plate Recognition (ANPR) capabilities into the wider CPMS ecosystem. ANPR technology enables vehicles to be identified automatically through camera feeds without requiring manual data entry by operators.

The long-term objective is to support a range of CPMS functions including:

- Automated vehicle entry and exit processing
- Parking session validation
- Barrier gate automation
- Vehicle whitelisting and blacklisting
- Enforcement and compliance monitoring
- Lost ticket management
- Cashless and account-based parking workflows
- Integration with payment and settlement systems
- Operational reporting and audit trails

Before ANPR functionality can be incorporated into production environments, a series of research and development tasks must be completed to assess technical feasibility, performance, accuracy, and integration requirements.

This document defines the initial R&D; problem set for engineers participating in the ANPR research programme.

Research Objective

Develop and evaluate a prototype ANPR pipeline capable of:

1. Detecting vehicle number plates from images.
2. Extracting plate regions accurately.
3. Recognising plate characters using OCR technology.
4. Validating recognised plate formats.
5. Producing structured outputs suitable for integration with CPMS services.

The outcome of this work will provide technical recommendations for future inclusion within the CPMS product roadmap.

Task 1 – Number Plate Detection Research

Objective

Develop a prototype capable of identifying the location of a vehicle registration plate within a captured image.

This capability forms the foundation of the ANPR workflow and will ultimately support automatic vehicle identification within CPMS entry and exit lanes.

Technical Scope

- Python
- OpenCV
- Haar Cascade or YOLO-based object detection

Input

Vehicle image captured from a camera source.

Example: car1.jpg

Expected Output

120 340 210 65

Where:

x = horizontal position

y = vertical position

width = plate width

height = plate height

Success Criteria

- Plate region is correctly identified.
- Detection accuracy is measurable.
- Bounding box coordinates are suitable for downstream OCR processing.

Task 2 – Plate Region Extraction

Objective

Following successful detection, extract the identified plate region into a separate image.

Input

- Source image
- Bounding box coordinates

Expected Output

output/plate_car1.jpg

Success Criteria

- Plate image is cleanly extracted.
- No clipping of characters occurs.
- Output quality remains suitable for OCR processing.

Task 3 – OCR Character Recognition

Objective

Develop an OCR module capable of extracting registration numbers from cropped plate images.

Technical Scope

- Tesseract OCR
- EasyOCR

Input

Cropped plate image.

Expected Output

BKT2931

Success Criteria

- Accurate recognition under normal lighting conditions.
- Consistent output formatting.
- Character confidence scores captured where available.

Task 4 – Visual Verification and Annotation

Objective

Generate annotated images showing the detected plate location.

Input

- Vehicle image
- Bounding box coordinates

Expected Output

output/detected_car1.jpg

Success Criteria

- Detection region is clearly visible.
- Annotation can be reviewed by engineering and operations teams.

Task 5 – Real-Time Camera Processing

Objective

Evaluate the feasibility of performing plate detection on live camera streams.

Input

Live webcam or IP camera stream.

Expected Output

Plate Detected

x=120 y=340 width=210 height=65

Success Criteria

- Stable processing performance.
- Near real-time detection capability.
- Acceptable resource consumption.

Task 6 – Multi-Vehicle Detection

Objective

Determine the ability of the ANPR solution to identify multiple vehicles within a single frame.

Expected Output

2

120 340 210 65

580 300 190 60

Success Criteria

- Accurate detection of multiple plates.
- Minimal false positives.

- Consistent performance across varying image conditions.

Task 7 – Registration Number Validation

Objective

Implement validation rules to assess whether OCR results conform to expected registration formats.

Input

OCR-generated registration number.

Example: BKT2931

Expected Output

VALID or INVALID

Success Criteria

- Invalid OCR results are identified.
- Formatting standards can be configured by region or customer deployment.

Final Deliverable

Develop a proof-of-concept ANPR service capable of producing a structured output suitable for CPMS integration.

Example:

```
{  
  "plate": "BKT2931",  
  "valid": true,  
  "bbox": [120,340,210,65],  
  "timestamp": "2026-05-29T10:15:22Z"  
}
```

The findings from this R&D; initiative will be used to assess the technical viability, implementation effort, infrastructure requirements, and business value of incorporating ANPR functionality into future CPMS releases.